Human factors evaluation techniques to aid understanding of virtual interfaces

R S Kalawsky, S T Bee and S P Nee

Advances in enabling technologies such as broadband wide area networks and the proliferation of the Internet has led to industry and home users looking beyond conventional communications media. Consequently, the telecommunications industry has been extending its application domain over recent years.Indeed, mediated communication has become a reasonably well established research area. Numerous modes of communication have been utilised successfully for various applications. From e-mail and text chat to videoconferencing systems, the use of mediated communication has become a part of daily life.

Virtual environments (VEs) are highly configurable media, ranging from relatively basic to extremely elaborate architectures. At their most complex, they promise a seamless interface between real and synthesised environments. The potential for media-rich environments is seeded in the technology's capacity to faithfully represent the participants (both physically and dynamically) and for those participants to present and interact with (shared) data in an intuitive manner. It is the latter of these attributes which is highly relevant to communications and is discussed in this paper.

The various configurations of virtual reality (VR) technology make matching the user to the technology an extremely complex task. Clearly, a basis for evaluating the effectiveness of these systems is required. Even though human factors (HF) evaluation and design techniques are well established in other human/computer interaction (HCI) fields, knowledge and understanding of virtual interfaces is limited. HF evaluation in VR is a complex subject and covers many aspects, such as basic human performance, cognition, and sensory capability. To address all these factors individually in an empirical fashion would demand a very long and expensive research programme. In addition, such studies may not predict the user's overall performance in a multi-modal VE. This paper deals with an alternative approach to understanding the issues of human performance in virtual environments via a process of top-down systems engineering evaluation. This paper is designed to provide an introduction to the assessment of virtual environments, and a reference for interface designers and researchers engaged in the investigation of mediated communication.

1. Introduction

It is clear that the capacity to convey meaning efficiently differentiates various types of media. Media richness is taken to refer to the capacity of the medium to communicate meaning. For example, text-based communication may be considered poor in terms of media richness since meaning is conveyed entirely in the explicit content of the message. Face-to-face communication may be considered 'information rich' with meaning conveyed in the message content and also in social cues such as facial and body gestures. Task demands may impose the need to represent shared data in the communication space. Clearly, a medium that exhibits the salient properties of face-to-face communication while providing an intuitive interface to a shared data space is required to enable a full range of communication interaction.

Human/computer interfaces are becoming more graphically oriented. The evolution of virtual interfaces or VR has led to an important new human/computer interaction medium with the potential to present the 'ideal' interface between the user and a synthetic computer-generated environment. VR offers an almost unlimited space in which to represent and interact in real time with data. The majority of virtual interfaces are designed to represent existing physical spaces or structures. However, VR systems are also very powerful for visualising and interacting with large or abstract multi-dimensional environments [1]. Unfortunately, the rapid pace of VR technology development has outstripped understanding of the underlying human factors issues, and solutions are being proposed without any real indication of the benefits (performance and cost) that such technology can afford.

Recently, there has been a move away from single participant VR (based on a head-mounted display) to group VR where several people simultaneously share the same experience either locally or via a remote network. This shift in delivery technology has been facilitated by the use of large screen semi-immersive VR systems. An important feature of a VR system, compared with other human computer interfaces, is its ability to create a sense of 'being in' a computer-generated environment. This is commonly referred to as achieving a sense of presence in the virtual environment. The term immersion is also used erroneously sometimes to describe the same experience, but here is taken to describe the extent of peripheral display imagery.

Zeltzer proposes a description of VR in his AIP cube [2] which sets out to define the components of a synthetic environment in terms of a co-ordinate system giving a measure of the quality of the system across three interacting parameters — autonomy, interaction and presence. Zeltzer's cube (Fig 1) illustrates the three different axes — defining a co-ordinate system which can be used as a qualitative measure of virtual environments. Autonomy is defined as the ability of the environment to act and react to simulated events. Interaction is the fidelity with which the environment deals with interactions between its participants both human and synthetic. Presence provides a rough (dimensionless) measure of the number and fidelity of available input and output channels. Zeltzer associates the position of an application within the cube with task performance. He indicates that while an evaluation of a VR system in these terms is highly task dependent, every design solution for a virtual environment can be characterised within these bounds.

At first sight, Zeltzer's AIP cube looks to be a good way of describing a particular VR system. However, it is very difficult to characterise a VR system in this way because of the lack of a clear definition for each axis. In particular, the term presence is difficult to specify in a simple way that would fit the AIP cube. It is tempting to try to classify attributes of a VR system in this way, and then devise a measurement process to suit, but there is a danger that the real performance-controlling factors of a VR system will not be addressed. Moreover, it is not easy to justify the use of a dimensionless performance parameter if it cannot be measured objectively or subjectively against clearly defined metrics.

To begin to understand how to evaluate the user's performance when using a VR system it is necessary to look further into the unique properties of the system. Traditional empirical evaluations, such as measuring the display resolution, are useful but do not necessarily relate to overall user performance. For example, it has been shown that performance in a VE is affected if one of the input modalities is removed [3]. This suggests that if we undertake empirically based evaluations we will not be able to draw the required conclusions regarding an integrated interface, rather we need to consider the complete VR system.

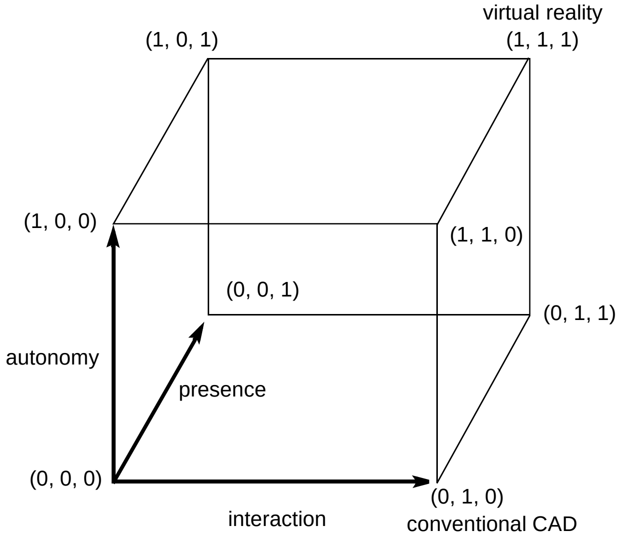

Fig 1 The AIP cube [2].

1.1 *Interaction in a virtual environment*

Research being conducted into interaction in a VE has largely been split into two areas. Firstly, there is the development of interaction devices and associated interaction aids. Buxton [4] illustrates a taxonomy of interaction devices that places input devices in a matrix according to the property sensed (rows), number of dimensions sensed (columns), requisite motor skills (sub-columns), and interaction directness (sub-rows). The main issue which arises from this representation is that peripherals tend to be designed with a specific set of tasks in mind (see Table 1).

The complex design issues for interaction devices have been highlighted by Wickens [5], including effects arising from specific device characteristics. For example, the number of dimensions offered by the interaction device may fall short of the task requirements. Different combinations of interaction dimensions can be employed in an attempt to overcome this problem. Perceptual issues associated with depth perception may be overcome by visual aids within the environment, or peripherals which permit stereoscopic viewing. Each of these methods has drawbacks, and performance degradation is often compounded by the introduction of perceptual aids and mapping solutions.

The second area of research in interaction in VEs does not match the perceptual structure of the interface to the human's low-level task requirements, but aims to extend the user's capabilities within the environment. This work aims to provide interaction metaphors which improve base task performance. For example, analysis of certain tasks may indicate that the user will be interacting with objects which are at widely varying distances. Rather than moving between the objects, a feature may be added which will allow the user to reach beyond arm's length. However, there

Table 1 A taxonomy of interaction devices [4].

		Number of dimensions							
		1		2				3	
Property sensed	Position	Rotary pot	Sliding pot	Tablet and puck Touch tablet	Tablet and stylus	Light pen Touch screen	Floating joystick	3D joystick	M T
	Motion	Continuous rotaty pot	Treadmill Ferinstat	Mouse			Trackball X/Y pad	3D trackball	M T
	Torque sensor						Isometric joystick		T

are various perceptual and motor issues which can affect performance when manipulating an object which is a significant distance from the user [6].

Virtual environments are often presented as ideal devices for procedural or perceptual motor skill training. However, as pointed out by Eggleston and Janson [7], Stytz [8] and Kalawsky [1], there is a need for a thorough investigation of the human factors of simulation fidelity.

> 'While VR systems have been around for some time (e.g., aircraft and automobile simulators), only recently has the capability existed to directly manipulate objects in a VE. Consequently, the data are generally scarce in addressing how properties of the VE effect [sic] human performance, ...' [7]

It has been shown that VEs need to run as smoothly and as fast as possible to increase presence [9] and decrease simulator sickness. This research leads to the conclusion that the designers of VEs need to employ a system resources versus virtual world sensory cue 'cost benefit analysis' of some description. Naive HCI decisions may prejudice the holistic performance of the system and lead to negative transfer in training applications.

This paper will endeavour to highlight both low-level design issues and the complicated issue of relating this work to a structured evaluation process. While subjective opinion is clearly valuable in a sometimes dynamic and highly interactive environment, the benchmark for performance must include an empirical representation of human activity in a dynamic task.

As in the design process, VR may borrow from previous work in other research areas. The application of tried and tested techniques will inevitably yield a formal description of the evaluation process at this level.

1.2 *Presence — the sense of 'being there' in a virtual environment*

An important feature of VR systems compared with other human/computer interfaces is their ability to create a sense of 'being in' the computer-generated environment. Other forms of media such as film and TV are also known to induce a sense of 'being in' the environment. Some authors have tended to use the term 'presence' to describe this effect. Unfortunately, the term presence has defied all attempts to define it in quantifiable terms. There have been numerous attempts to produce a single metric representing the degree of presence for a VE and then relate this to some measure of human performance [9]. Traditional evaluation techniques do not take into account attributes such as presence and greater interactivity. Presence is a multi-dimensional factor that is highly dependent upon a number of inter-related factors. Therefore, understanding presence and its components could make an important contribution to the future development of effective VEs. It is claimed by Sheridan [10] that the degree of presence is enhanced by being able to interact with a VE. This interaction is not confined to the two or three degrees of freedom that conventional computer interfaces provide, increasing the number of factors affecting the sense of presence. Measurement of user performance in a VE typically involves dealing with a range of measures, including objective measures (task demands, task results and correlated measures — error numbers, achieved task levels, etc), subjective measures (on-line evaluations, post-test evaluation questionnaires, explanation of high stress events), psycho-physical measures (detection of the stimulus, recognition of the stimulus, stimuli discrimination, magnitude), physiological measures (heart rate, blood pressure, respiration rate, ECG), task performance, and learning efficiency. Unfortunately, presence does not have a single physical manifestation that can be measured objectively. This means that a potentially more useful research goal is the development of a functional representation of presence. A simplified parametric equation for presence, ρ, can be summarised as:

$$\rho = (\alpha; \beta; \chi; \beta_d; C; SA, \Delta SA; \gamma; \mu; \tau, \Delta\tau; \Psi; I_{qty}; I_{qual}; T_e; \kappa; \sigma; \theta; \phi; d_m; t, \Delta t$$

The factors and functions are given in Table 2.

This parametric equation does not take into account every factor that relates to the subjective feeling of being in a VE. However, by assuming that a VE is required to communicate information to a user, who then is required to perform a series of tasks, it is reasonable to look at the consequential demand and supply on attentional resources. It is thus possible to make comparisons against task performance. Where the user is required to assimilate the information and understand the situation it is possible to begin to explore the user's situation awareness. An overview of a technique currently under development which is hoped to identify the dominant factors that lead to a sense of presence is given in section 5.2.

2. The case for formal HF performance evaluation

In order to provide an effective VE we must understand the user in terms of interaction and cognition. However, there have been surprisingly few attempts to thoroughly understand human interaction with VEs. The majority of the research and development to date has predominantly focused on the enabling technologies. Consequently, interfaces have emerged that do not necessarily fulfil the designer's expectations. Their resulting design represents a less than satisfactory interface which then impedes the user — who tends to prefer more traditional methods of interacting with a computer. It is not surprising that this situation has emerged because the technology is seductive and understanding of human performance and capability is underdeveloped. If successful VR-based products are to be achieved, then we must invest in HF-based evaluations for which there are two perspectives. Firstly, there is an evaluation that is human centred and addresses human performance capabilities, and secondly there are technology issues that relate to how the system meets the requirements of the user.

2.1 Objective performance measurement

Objective measures are generally preferred when performing an HF analysis as they provide a measure of performance against quantifiable parameters. Such parameters include number of errors made, time to complete task, etc. Objective measures are usually collected on-line and without user intervention. The advantage is that the user is not subjected to questioning that could interfere with the outcome of the trial. Objective measures of human performance are well established for traditional manual control tasks. For example, piloting aircraft and driving automobiles have provided the motivation for much of the work in this area [5, 11]. There is a clear relationship between the mechanics of flying or driving through a real-world environment and moving the user's viewpoint through a VE. However, the evaluation techniques used by researchers in this area need not be confined to egocentric motion. Indeed, target acquisition and object manipulation may also be evaluated with these techniques [12, 13]. The

Table 2 Salient parameters in the definition of presence.

Factor	Function
Demand of attentional resources	α = demand on attentional resource
Supply of attentional resources	β = supply of attentional resource χ = concentration of attention β_d = division of attention C = spare mental capacity
Understanding of situation	SA, ΔSA = situation awareness and change in situation awareness γ = understanding of situation μ = complexity of situation τ, $\Delta\tau$ = spatial awareness and change in spatial awareness Ψ = familiarity of situation
Information	I_{qty} = information quantity I_{qual} = information quality T_e = elapsed time in environment
Technological factors	κ = sensory modality σ = degree of immersion θ = field of view subtended by the participant's eye ϕ = field of regard of participant d_m = display mode (binocular, monocular) t = update rate Δt = time lag (propagation delay between event and consequential action)

application of these techniques to VEs is immature. Let us consider the fundamental control requirements for effective interaction in a VE. Table 3 shows the bottom level motor tasks for a virtual interface and the associated user requirements.

Certain applications require more complex high-level interactions. For example, object modelling tools enable the user to scale objects interactively. This task may be broken down into the selection and manipulation (typically translation) of control points. Therefore, the actions highlighted above can be seen to form the baseline for a generic VE control interface.

Techniques for objective evaluation of control components have been split between static and dynamic tasks. The static tasks tend to take the form of a docking procedure, where the user is required to match a cursor and target object as quickly and accurately as possible, [14—16]. Statistical significance can be obtained with this technique by repeating the process a number of times with a pseudo-randomly placed target. The experimenter would typically record the time to complete for each trial and accuracy. The difficulty in implementing this process is in whether to emphasise accuracy or speed of completion when describing the task to the user. Maybe as a result, this technique has occasionally failed to produce significant results when comparing interface components [14]. In this case subjective preference proved to be a more reliable technique for comparison.

The dynamic task usually involves the user tracking a target object which is moving in an unpredictable manner [5, 11—13, 17, 18]. The dynamic task is continuously monitored over a period of a few minutes (Poulton [11] suggests between 1 and 4 minutes, per trial). From the data relating to cursor and target state (e.g. position and/or orientation) accrued over the duration of the track, a number of different types of analyses may be carried out. Poulton gives a comprehensive review of recommended and not recommended methods of scoring. Much of the work in VEs has used RMS error as an overall indication of performance. While this tends to be reasonably reliable when applied over a number of tracks, it is a gross measure of performance and tells us nothing of the subtleties of how the interface was being used. A more detailed analysis may be carried out in the frequency domain [5, 11, 12, 19]. A further measure of noise (or remnant) indicates where the user was making movements which were not present in the target's movement. Analysis in the frequency domain is less common in VE evaluation, because the technique has tended to be related to single-dimension tracks, where VE interfaces usually deal with multi-dimensional tracks. The demand on attentional resources will tend to be higher than with a static task. Clearly where the dynamic task may be suitable for representing dynamic situations, such as in a flight simulator, the static task may be more representative of a CAD system where the attentional demands are lower. It is important at this stage to consider what defines an ideal control interface. Wickens [5] highlights three criteria for ideal control (see Table 4).

The key to effective objective performance monitoring for VEs is an understanding of the task requirements of the specific VE application. This will determine the emphasis placed on each of the control components and consequently the relative importance of each of the interface components. The traditional human/computer interaction field provides a number of techniques for breaking down task requirements for simple computer-based tasks and analysing human performance at a cognitive and motor level within this framework (this approach is exemplified by Card, Moran and Newell [20]). As yet, this type of analysis of 2-D interfaces does not appear to have been comprehensively extended to dynamic 3-D VEs. This may be due to the complexities, both cognitive and perceptual, of a VE task. Instead, researchers have concentrated on evaluating human performance with the interface components (as highlighted above) and using subjective assessment for the completed interface.

2.2 *Subjective performance measurement*

In HF evaluations it is not always possible or desirable to run experiments where objective performance data is obtained. For instance, in complex experiments it can be very difficult to fully instrument the system under evaluation. Instead, measures that reflect subjective opinion have to be collected from trial participants (see Table 5). To the engineer, analysis of subjective data is viewed as a very imprecise process. Psychologically based research methods can provide data on aspects of human performance.

Table 3 Low-level motor tasks for a VE.

	Mode	Requirement
Object manipulation	Selection	The user must be able to indicate objects of interest both for inspection and for manipulation.
	Translation	The user will tend to require the ability to move objects within the environment.
	Rotation	The user will require the ability to orientate objects within the environment. In many instances the translation and rotation aspects of object manipulation will be coupled.
Egocentric motion		The user must be able to move freely through the environment.

Table 4 Control criteria for ideal control from Wickens (1986).

Control criteria	Description
Low error	The differences between target and cursor are recorded over time. When considering a tracking task the error may arise in a number of different ways. There may be an error in phase and/or gain. Also there is likely to be a certain amount of noise in the system. RMS error analysis is used to establish a gross measure of error.
Stability	This refers to the user's ability to achieve a bounded response to the target movements. The contrary would cause oscillatory behaviour, where the user could not home in on the movement of the target. Stability is measured through linear systems analysis. Typically the experimenter would use a quasi-linear model to represent the subject's response with the controller [21, 22].
Control activity	Wickens [5] states: "In many circumstances performance is 'better' if the amount of control activity delivered to the system input ... is low."

Table 5 Examples of subjective workload.

Type	Measures
Workload	Amount of perceived mental and physical effort required to complete task
Attentional resources	Amount of attention devoted to particular parts of the task
Task difficulty	Perceived view of the level of difficulty to achieving task
Understanding	Degree of understanding of the requirements of the task

However, to be useful a large number of subjective responses need to be collected in order to achieve statistical significance. Particular care must be taken to ensure that the participant pool is representative of the eventual user group for the VR system. A typical method of obtaining subjective data is via the use of carefully designed questionnaires, attitude scales, and surveys that test current opinions or patterns of behaviour. The temptation to encourage the user to provide verbal commentary on their actions should be discouraged unless it is known that the additional verbal task will not interfere with the principal task. Kinsbourne and Cook [23] determined that concurrent speaking has an interfering effect on manual tasks. They explained the effect by the fact that verbalisation and right-handed motor activity are controlled from the left cerebral hemisphere, whereas left-handed motor performance is controlled by the right cerebral hemisphere. Similar results were obtained by Hicks et al [24].

Evaluation of VR system usability is important for identifying and subsequently improving the user interface. Usability is a multi-dimensional term that relates to different aspects of a product. Shackel [25] stated that: 'The usability of a computer is measured by how easily and how effectively the computer can be used by a specific set of users, given particular kinds of support, to carry out a fixed set of tasks, in a defined set of environments'. By controlling the usability engineering process it is possible to achieve the following [26]:

- defining usability goals through metrics,
- setting planned levels of usability that need to be achieved,
- analysing the impact of possible design solutions,
- incorporating user-derived feedback in product design,
- iterating through the 'design-evaluate-design' loop until the planned levels are achieved.

The complexities of the VR system make it difficult to apply traditional tools in the evaluation process. Usability evaluation is a useful technique in the designers' tool-box, although it should be noted that the interface attributes are interdependent. In addition, performance and subjective measures are sensitive to different task factors.

3. Issues of evaluation

The environment, personal capabilities, individual motivation and the tasks to be performed govern the user's performance with a virtual interface. Also, during the evaluation process the user will experience fewer sensory cues in a VE than in the real world. This inevitably means that mental models relating to the real world may only partially fit the case for VEs.

If a user is performing a task in a VE, it is possible that an experimental evaluator (on the outside) will interfere with the performance of the user. This type of problem can only be overcome by careful experimental design. In situations where highly realistic virtual environments are being used, the lack of certain real or redundant sensory cues may have a detrimental effect on the user's performance and subjective experience. An equally important issue is one of perceptual conflict where for example, dominant visual cues may conflict with whole body kinesthetic cues. This is known to have been the cause of accidents in the aerospace sector where pilots have tended to believe their own proprioceptive senses rather than aircraft instrumentation. In some instances it will not be possible to avoid such complexities, as the enabling technology will be limited. Under these circumstances an understanding of empirical human performance becomes important.

4. Examples of techniques in use

It is not possible to describe the full range of HF techniques applicable to VE evaluation in this paper. However, a few of the techniques being used by the Advanced VR Research Centre (AVRRC) at Loughborough University are reviewed below with a summary of the results that have been obtained.

4.1 Objective performance measurement examples

4.1.1 Evaluation of control interfaces for virtual environments

One approach for the evaluation of control interfaces for VE gradually builds the interface from its components, testing performance at each stage. An example of this technique is described below. Figure 2 illustrates an experiment adapted from Chen et al's study for rotation controllers [14]. Chen used separate displays for the target and cursor. Here the target and cursor are amalgamated into one display. This representation has two advantages; it is easier for the subject to determine when the two objects match, and in a dynamic task the subject's visual attention is not distracted to another part of the screen.

A number of rotation controllers were compared with static and dynamic matching tasks. A repeated measures analysis of variance (ANOVA) was used to test for statistical significance. Although there proved to be no significant difference in performance across controllers in the static matching task (time to complete, accuracy), a difference was shown across controllers in the dynamic task (RMS error).

Figure 3 illustrates the next stage in the comparison of VE interfaces for object manipulation. A number of controllers and peripherals were compared using static and dynamic matching tasks in an environment where the target was allowed to rotate and translate. Where the rotation controllers from the rotation study were incorporated into the interface, a training session using the environment from the rotation study was implemented. Here the static and dynamic tasks showed no difference in performance between rotation controllers, which indicated that, while one controller may have proven more efficient when considered in a purely rotation-based task, the attentional demands required by the controller meant that performance was degraded so as to show no advantages in a dual-task environment. This finding was supported by reported subjective impressions of the controllers.

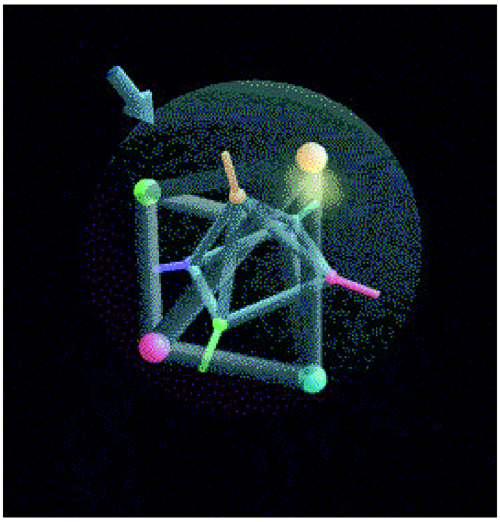

Fig 2 Virtual sphere rotation controller.

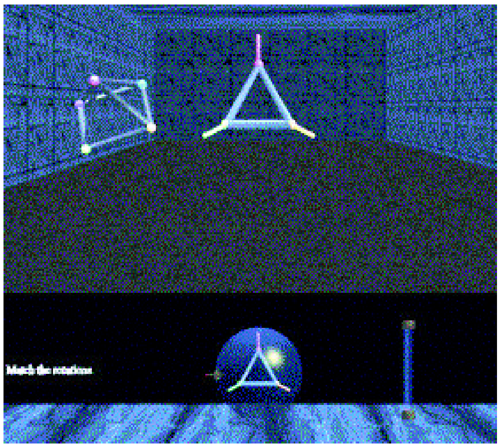

Fig 3 Object manipulation in a virtual environment.

This approach of testing the interface at each stage will enable the designer to build a clear picture of how the component parts of the interface impact on one another. At a higher level some form of task analysis with respect to the final application will enable the designer to develop representative tasks.

4.1.2 Comparison of dynamic task performance in real and virtual environments

Another approach involves the comparison of human performance in a real environment with human performance in a VE. By using a suite of established human performance models such as Fitts Law [27] in a comparative study, valuable insight is gained into human motor performance in VEs. In this study, behavioural, performance and error isomorphisms with the real world are identified with their factors and determinants. The rationale for this approach is that some VEs have a requirement to 'mimic' reality closely.

The aim of this research is to identify established human performance models and to test their validity within a task-specific VE. By using task-based VEs the impact of design and system decisions can be observed as a variance of human behaviour from the real-world model. The correlation between a virtual/real-world task pair is then tested for significant differences due to environment design. Unlike other measures, the test of quality is not self-referential with regard to the VE, but based on comparison with real-world performance. One experiment to assess the feasibility of this comparative evaluation process was a Fitts' reciprocal tapping task. Fitts' law is a successful model of human psychomotor behaviour. It has been used to determine the effectiveness of interaction devices in many studies [28, 29]. The devices that have been studied range from computer mice to foot-pedals to virtual reality gloves. Interestingly, Fitts' law relies on a theorem of information processing (Shannon's Theorem) and not a theorem of human movement. Equation (1) gives the fundamental form of the law:

$$MT = a + b \cdot \log_2\left(\frac{2A}{W}\right) \qquad \text{...... (1)}$$

where MT is the time to target, A is distance to target, W is the size of the target and a, b are determined by linear regression. The independent variable of interest is the index of difficulty (ID) and is shown in equation (2):

$$ID = \log_2\left(\frac{2A}{W}\right) \qquad \text{...... (2)}$$

The final term of interest in Fitts' law is the index of performance (IP):

$$IP = \frac{ID}{MT} \qquad \text{...... (3)}$$

From these equations calculating the time to target (MT) over differing index of difficulty (ID) allows us to form a regression line with slope and intercept coefficients. The inverse of the slope of the regression line is equal to the index of performance (IP) and our measure of the performance of a device. The baseline of performance used in this experiment is based on the interpretation that devices interacting with identical systems leading to higher IPs are more effective. The method used here to compare performance in a VE with real-world performance involves collecting on-line data and correlating between the ID and MT of a user tapping between two virtual plates. In the first experiment the factors were the presence of a retinal disparity cue, and the modality of the stimulus to indicate a collision of probe with plate (see Fig 4).

From the collected data it is possible to correlate between ID and MT and compare them with real-world results to examine the participant's adherence to the model. The correlation coefficient collected and its closeness to the real-world correlation coefficient are proposed as candidate measures of the behavioural isomorphism of the environment for the task. Further experiments are under way which extend the task set.

4.2 *Subjective performance measures*

Subjective performance measures are very important in collecting personal opinion from users. Careful use of subjective techniques can reveal important aspects of the system design that may be of immediate use to the designer. The main benefit of subjective techniques is the ease with which they can be administered to users, provided a carefully designed experimental procedure has been devised to remove experimental biases. A range of subjective performance measurement techniques are introduced below.

4.2.1 Usability evaluation

With the growing interest in VR-based applications there is an urgent requirement to quantify the benefits of potentially costly VR technology. When people try out VR

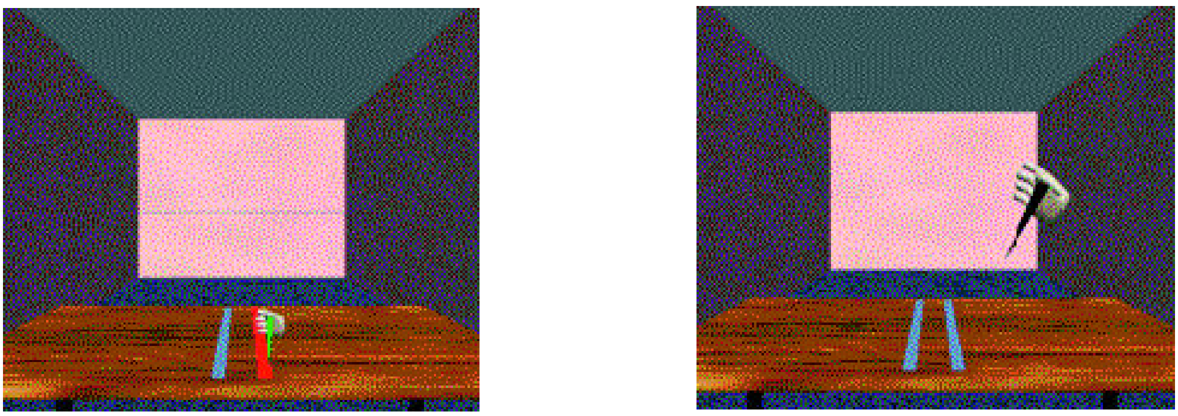

Fig 4 Showing the virtual hand and probe in 'collision' and 'non-collision' modes.

systems they often complain about a whole variety of aspects of the interface, from display resolution through to the method of interaction. Such feedback is not always helpful if it is received in an unstructured manner. For example, when someone complains that the display updates in jerks, the natural assumption is that the computer cannot update the display quickly enough. However, a more detailed system analysis will show that this problem could have arisen from a number of areas in the system, such as the head-tracking system, the graphics computer, a poorly designed model, an inappropriate simulation, or even the user interaction device. The user may not know how the VR system is designed, but their perception of the usability of the system is vital if effective VR systems are to be developed.

VRUSE was designed as a ten-part questionnaire where each part addresses a key usability factor of the interface. The ten factors were derived from a number of iterations around an item/reliability analysis conducted on an initial set of nearly 250 questions. The goal for each of the ten usability factors was derived from discussion with other VR experts and users. The item/reliability analysis reduced the number of questions down to a more manageable 100 (see Table 6).

Example application of VRUSE — a presentation of a complete usability evaluation of a VR system is outside the scope of this paper because a comprehensive usability analysis requires consideration of the context of use. A full usability evaluation will typically involve collection of data as illustrated in Table 6. Loughborough University routinely collects this data in most of its VR usability evaluations since it provides full diagnostic information about a given VR system. To illustrate the potential of VRUSE, part of a typical evaluation conducted by the AVRRC is now reported.

The evaluation involved using a fully immersive VR system (head-mounted display) to create, link and animate a series of objects while in the virtual environment. The evaluation is part of a larger study to determine if people perform better in spatially immersive environments compared with non-immersive environments. A within-subjects experimental design was used. Nineteen participants were selected who had experience with computer systems, were right-handed and had normal or corrected-to-normal vision. Prior to the formal evaluation a training session was undertaken to ensure all participants reached a certain level of proficiency.

A fully immersive VR system (Division's PV100 comprising head-mounted display, Polhemus Fastrak, Division 3D Mouse and dVS version 3.1.2) was used to provide the virtual environment. The immersive environment provided a virtual tool-box that could be used to create instances of objects and the means to link objects together to create object hierarchies. The tool-box also facilitated the process of animating objects so that a dynamic scene could be created. During the trial each

Table 6 Key usability factors and goals.

Usability factor	Goal
Part 1 Functionality	The interface should be able to provide the level of functionality (control) the user expects in order to complete a task.
Part 2 User input	The user should be able to interact with and control the virtual environment in a natural manner.
Part 3 System output (display)	Information displayed to the user should be understood, unambiguous and necessary to complete the task.
Part 4 User guidance and help	The user should be able to request help via on-line assistance.
Part 5 Consistency	The operation of a VR system shouild be consistent with the user's understanding and convention.
Part 6 Flexibility	The VR system should not constrain the user who should be able to interact with the system in a flexible manner.
Part 7 Simulation fidelity	In order to be useful a VR system needs an underlying model or simulation to control the virtual environment.
Part 8 Error correction/handling and robustness	All computer systems should provide error correction and recovery before a permanent change is made.
Part 9 Sense of immersion/presence	A VR system should allow a user to feel part of (or immersed in) a virtual environment.
Part 10 Overall system usability	Overall a VR system should be intuitive and easy to use.

participant was required to perform a series of create, link and animation operations according to a predefined task list. The results illustrate the potential of VRUSE. Analysis of the results considered the correlation between the factors of simulation fidelity and the sense of immersion/presence.

The participants' belief that the quality of the simulation enhanced performance was correlated with various factors, including a feeling of being present in the environment, a sense of scale, and a belief that the field of view enhanced performance. The finding that the quality of simulation increased performance was highly correlated with the sense of presence, supporting the case for using fully immersive VR systems in certain applications. This result was expected because the task in the experiment involved manoeuvring within the virtual environment and placing objects in specific locations and a good spatial awareness (produced by the field of view of the head-mounted display) was essential for maintaining orientation cues.

Although participants tended to report that accurate scaling of the environment was correlated with a sense of presence and improved performance, the designer should be cautious in ensuring they understand the significance of this result. If a virtual environment is built to model a real world situation then the user's *a priori* knowledge may lead to better performance if the virtual environment has the same scale factor. It should be noted that situations may occur where a scaled virtual environment is more sensible. VRUSE is good at highlighting where performance may be compromised by a non-obvious scaling factor.

It was interesting to note that the participants felt that they had the right level of control which was correlated with a good sense of scale and a sense of immersion. All the principal determinants for the sense of presence suggested by Sheridan [10], (extent of sensory information, ability of observer to modify their viewpoints, ability to modify spatial relationships of objects) and Kalawsky [1] (closed loop performance due to an operator-induced motor movement) are consistent with the results obtained by VRUSE.

Analysis of the questionnaire data reveals important diagnostic data that can be used by a system designer/developer to pin-point problematic aspects of the VR system's interface. The questionnaire is able to discriminate between different usability aspects of an interface such as user input, display output, simulation fidelity, etc. It is quite easy to correlate this information with other performance data such as subjective workload and objective task performance.

4.2.2 VR situation awareness rating technique (VRSART)

The process of collecting situation awareness (SA) data can have an interfering influence on a task if the data is collected during the trial. Early SA rating techniques required the experimenter to stop the experiment at defined points and run an SA debriefing questionnaire. Unfortunately, this approach modifies overall task performance because the participant's cognitive loading is modified as a result of the interference. Instead, it has been found by experimentation that SA debriefing is more effectively run immediately after the trial. A reliable and robust method of eliciting SA measures is under investigation at Loughborough. The objectives behind the method tool are:

- to provide a sensitive computer-based diagnostic aid to assist in relating task performance to the contributory factors that lead to a sense of presence,
- to provide a structured method of determining the impact of presence on the efficiency of a spatially immersive VR system,
- to partition presence factors into specific categories,
- to be a sensitive indicator of problematical areas of the VR user interface,
- to provide immediate feedback on user performance criteria.

VRSART has been extended to address the correlation between demand/supply of attentional resources on situation awareness and the sense of presence. Although task performance is of primary concern the impact of secondary factors is also extremely important because it may be possible to tailor the virtual environment to enhance situation awareness with a consequential improvement in overall task performance. Phase 2 VRSART deals with six principal factors, as given in Table 7.

A special computerised questionnaire was developed so that subjective data relating to the factors shown in Table 7 can be collected immediately after a trial and with minimum influence on the participant's subjective ratings. Other data relating to technology factors (update rate, resolution, field of view, etc) are also collected.

VRSART in the context of an evaluation — VRSART must be administered immediately after the trial and before any subjective workload or usability tests have been conducted. Ideally, the participant should complete VRSART before leaving the experimental facility. A trial was undertaken to compare the differences between a task performed using a desk-top VR system compared to a semi-immersive projected VR system. Care was taken to ensure that the effective field of view between the two systems was matched. The same interaction devices were used, in this case a conventional mouse. Participants were given a set of instructions and a paper map of part of an industrial processing plant. The initial task was to familiarise themselves with the spatial location of various objects. Prior to the trial a period of training and system familiarisation

Table 7 Principal factors used in VRSART.

Factor	Description
Demand	Demand on attentional resources — what percentage of resource was being demanded. Unpredictability of situation — to what extent was the situation unpredictable. Variability of influencing variables — an indication of how many variables were influencing the situation.
Supply	Supply of attentional resources — what percentage of available resource is being supplied. Degree of readiness — readiness to deal with the situation. Concentration of attention — an indication of mental effort expended to deal with situation. Division of attention — percentage of attention time devoted to dealing with situation. Spare mental capacity — an indication of how much spare mental capacity is left to deal with additional tasks.
Information	Quantity of information — how much content is being received. Quality of information — level of excellence of information being received.
Understanding	Understanding of situation — rating of degree of understanding. Previous experience — rating of familiarity of situation. Situation awareness — perception of complete awareness of situation. Complexity of situation — expression of how complicated the situation was, familiarity with situation, or previous experience. I felt isolated and not part of the virtual environment. I felt disorientated in the virtual environment. I had a good sense of scale in the virtual environment.
Presence	I got a sense of presence (i.e. being there).
Technology	The quality of the image reduced my feeling of presence. I felt a sence of being immersed in the virtual environment. The display resolution reduced my sense of immersion.

was given. Twenty-four participants were selected and half the group experienced the desk-top system first followed by the semi-immersive system. The other half experienced the systems in the reverse order. The primary task involved navigating around the virtual environment, moving to specific locations and then performing a number of disassembly tasks. Finally, the participants had to negotiate their way back out of the industrial plant to the original starting point. Immediately after the trial, VRSART was administered, followed by VRUSE. Continuous video recordings were made during the trial and then analysed off-line to determine specific completion times for previously defined tasks (see Fig 5).

As with any subjective measurement tool, care must be taken with interpretation of the results. While it is possible to employ VRSART by itself it is much better to use it as part of an overall evaluation programme where other user performance data (typically, usability, task workload, context analysis, task performance, etc) is collected. This makes it easier to relate any unexpected results with other quantitative data.

4.3 The role of the perceptual model

As the number of human factors experiments involving VR systems increases, we need a way of collating the data. The multi-modal nature of a virtual interface is beneficial but complex. Its complexity arises from the wide variation of interface technology that the system can support. Thus it becomes difficult to specify the best interface for a particular task. Clearly, HF evaluations must be undertaken with respect to the multi-modal nature of the interface, but it remains difficult to relate these results to a variety of situations because of the complex interrelation between the data.

Due to the complexity of dealing with multi-modal interfaces and the time it takes to perform human factors based experiments, there is a growing requirement for the development of multi-modal perceptual models. Such models would allow rapid iteration of different design solutions in an attempt to produce a better system for evaluation. Single sensory perceptual models exist but are unable to deal with multi-modal interaction. In a VR system the integration of multi-sensory data is important if a user is to profit from enhanced performance.

A perceptual model as proposed by Hollier and Voecker [30] would greatly aid our understanding of how and where to optimise any virtual environment. The creation of a reliable perceptual model is an involved process which relies on collecting large quantities of data and developing relationships between the data. Testing and validation of the model requires comparison of model-predicted performance against experimental evidence. The quality of the perceptual models under development is gradually evolving over time.

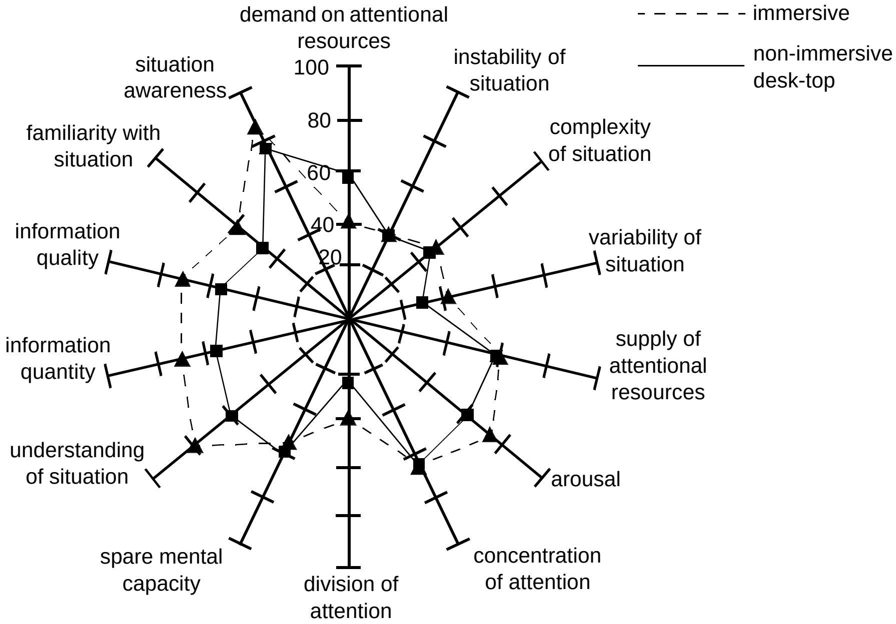

Fig 5 An example of the output from VRSART.

5. Conclusions

The evaluation of virtual interfaces is still in its infancy compared to oth er traditional computer interfaces. The challenge is dealing with an interface where the designer has total flexibility in the design. The potential for multi-sensory interfaces presents additional complexity. Perhaps the greatest challenge lies in the trade-off between different performance requirements across the different modalities. A top-down approach may yield some convergence in our understanding, in preference to treating the interface as presenting separate interface modalities.

HF experiments have shown the impact of factors such as display resolution on task performance. While this information is important from an academic perspective, industrial developers are also interested in higher order HF issues such as the trade-off of a range of parameters against overall user performance. This different perspective is necessary so that developers can optimise their products to meet the needs of the end user. Key questions are frequently asked such as: “What spatial resolution is required at a given update rate to complete a given task without compromising overall task performance?” Armed with this knowledge the system developer would be able to adjust the performance budget without an over expensive solution. Having derived the required level of system performance it is possible to look at various human factor trade-offs to fine-tune the system design. The idea of a top-down HF systems view is not new and has been successfully used in the aerospace industry for decades. HF is frequently overlooked leading to products that are difficult to use. VR is a rapidly emerging area of development, and it is therefore no surprise that various products exhibit problems which must be removed if the concept is to be more widely adopted.

The rapid pace of technological development is outstripping our current knowledge and we will continually have to refine our models of human performance in VEs. The range and diversity of all aspects of HF prevents us from undertaking a complete systematic analysis of all combinations of interface components. By taking a top-down approach it is possible to yield a substantial amount of useful information, examining the most critical issues first. The ultimate quest is for a set of robust evaluation methodologies and metrics that support our required level of understanding.

Acknowledgements

This work was supported in part by EPSRC and the Joint Information Systems Committee whose support is gratefully appreciated. Support from commercial organisations and, in particular, BT, British Aerospace PLC, Division Inc, MultiGen Inc, SEOS Displays Ltd, Trimension Ltd and Silicon Graphics Inc, is gratefully acknowledged.

References

1 Kalawsky R S: ‘The science of virtual reality and virtual environments’, Addison-Wesley-Longman (1993).

2 Zelzter D: ‘Autonomy, interaction and presence’, Presence, 1, No 1, pp 127—132 (1994).

3 Pausch R, Shackleford M A, Proffitt D: ‘A user study comparing head-mounted and stationary displays’, Proc IEEE Symposium on Research Frontiers in Virtual Reality (1993).

4 Buxton W: 'Lexical and pragmatic considerations of input structures', Computer Graphics, 17, No 1, 31—37 (1983).

5 Wickens C D: 'The effects of control dynamics on performance', in Boff K and Kaufman A (Eds): 'Handbook of perception and performance', pp 39.1—39.60 (1986).

6 Poupyrev I, Weghorst S, Billinghurst M, Ichikawa T: 'A framework and testbed for studying manipulation techniques for immersive VR', Proceedings of VRST'97, ACM, pp 21—28 (1997).

7 Eggleston R G and Janson W P: 'Field of view effects on a direct manipulation task in a virtual environment', Proceedings of the Human Factors and Ergonomics Society 41st Annual Meeting, pp 1244—1248 (1997).

8 Stytz M: 'Distributed virtual environments', IEEE Computer Graphics and Applications, 16, pp 19—31 (1996).

9 Slater M, Usoh M and Steed A: 'Depth of presence in virtual environments', Presence, 3, pp 130—144 (1994).

10 Sheridan T B: 'Musings on telepresence and virtual presence', Presence, 1, No 1, pp 120—125 (1992).

11 Poulton E C: 'Tracking skill and manual control', Academic, London (1974).

12 Jagacinski R J, Plamondon B D, and Miller R A: 'Describing movement control at two levels of abstraction', in P A Hancock (Ed): 'Human factors psychology', pp 199—247 (1987).

13 Zhai S, Buxton W and Milgram P: 'The Silk Cursor: investigating transparency for 3D target acquisition', in CHI'94, pp 459—464 (1994).

14 Chen M, Mountford S J and Sellen A: 'A study in interactive 3-D rotation using 2-D control devices', Computer Graphics, 22, No 4, pp 121—129 (1988).

15 Liu A M: 'Depth cues for telerobotic tasks in virtual environments', PhD thesis, University of California, Berkley (1992).

16 Zhai S: 'Human performance in six degrees of freedom input control', PhD Thesis, University of Toronto (1992).

17 Hancock P A: 'Effects of control order, augmented feedback, input device and practice on teaching performance and perceivec workload', Ergonomics, 39, No 9, pp 1146—1162 (1996).

18 Knight J L: 'Manual control and tracking' in Salvendy G (Ed): 'Handbook of human factors', pp 182—218 (1987).

19 Rouse W B: 'Estimation and control theory: application to modelling human behaviour', Human Factors, 19, No 4, pp 315—329 (1977).

20 Card S K Moran T P and Newell A: 'The psychology of human-computer interaction', Lawrence Erlbaum Associates, Hillsdale, NJ (1983).

21 McRuer D K and Krendel E S: 'Dynamic response of human operators', Journal of the Franklin Institute, 267, pp 381—403 (1959).

22 McRuer D K and Krendel E S: 'The human operator as a servo system element', Journal of the Franklin Institute, 267, pp 511—536 (1959).

23 Kinsbourne M, and Cook J: 'Generalised and laterialised effects of concurrent verbalisation on a unimanual skill', The Quarterly Journal of Experimental Psychology, 23, pp 341—345 (1973).

24 Hicks R E, Bradshaw G J, Kinsbourne M and Feigin D S: 'Vocal-manual trade-offs in hemispheric sharing of human performance control', Journal of Motor Behavior, 10, pp 1—6 (1978).

25 Shackel B: 'The concept of usability', Proceedings of IBM Software and Information Usability Symposium, Poughkeepsie, NY, pp 1—30 (September 1981).

26 Preece J et al: 'Human computer interaction', Addison-Wesley (1994).

27 Fitts P M: 'The information capacity of the human motor system in controlling the amplitude of movement', Journal of Experimental Psychology, 47, pp 381—391 (1954).

28 MacKenzie S I: 'Fitts' Law as a research and design tool in human-computer interaction' Human-Computer Interaction, 7, pp 91—139 (1992).

29 Plamondon R and Alimi A M: 'Speed/accuracy trade-offs in target-directed movements', Behavioural and Brain Sciences, 20, pp 279—349 (1997).

30 Hollier M P, Voelcker R: 'Towards a multi-modal perceptual model', BT Techol J, 15, No 4, pp 162—171 (October 1997).

Roy Kalawsky spent 18 years working for British Aerospace where he was responsible for cockpit research across the Military Aircraft Division. He has undertaken research into virtual interfaces since 1978 and has led several international collaborative human factors based projects involving virtual interfaces. In 1995 he was appointed Professor of Human-Computer Integration at Loughborough University where he has established the Advanced VR Research Centre. The centre is involved in the development and improvement of virtual interfaces for a wide range of application areas.

Simon Bee received a BSc (Hons) in Computer Science from Manchester University. On graduating he joined British Aerospace Military Aircraft Division, where he delivered a number of high profile VR projects. In 1995 he commenced a PhD at Loughborough University. His doctoral thesis is entitled ‘Evaluation of Control Techniques in Virtual Environments’. In September 1998 he joined the Understanding Virtual Environments group at BT Laboratories. His current research interests include mediated communication, group dynamics, virtual reality and human performance.

Simon P Nee is in the final year of his Doctoral studies at Loughborough University. He received his BSc in Physics and Computing from Oxford Brookes University in 1995.

He joined the AVRRC in 1996 after gaining financial sponsorship for his studies from British Aerospace PLC. Among his research interests are behavioural isomorphism in virtual environments, simulation fidelity, and the use of virtual environments in the classroom.